

Assignment Two (2)

1. Title and Collection Method:

Dataset: Identifying Factors that influence student success.

Description: this dataset is about understanding which factors affect whether student passes or fail in university or school.

Collection Method: I have collected this dataset by myself using questionnaire via Google forms and questions asked about study habits and lifestyle.

2. Features & Labels:

I have used for this dataset five features and two label detailed bellow.

Features (X)

-
- Study per day
 - Attendance Rate
 - Sleep per night
 - Time on social media
 - Parental support

Label (Y)

-
- Pass
 - Fail

3. Dataset Structure:

- Rows: 52 students.
- Columns: 7 (5 - Features and 2 – Labels)

Study_Per_Day	Attendance_Rate	Sleep_pN	Time_SM	Parental_S	Result
3–4 hours	70%–90%	5–6 hours	3–4 hours	Always	Yes
Less than 1 hour	50%–70%	5–6 hours	More than 4 hours	Occasionally	No
More than 4 hours	Above 90%	More than 8 hours	3–4 hours	Always	Yes

Assignment Two (2)

3–4 hours	70%–90%	7–8 hours	3–4 hours	Occasionally	Yes
1–2 hours	70%–90%	7–8 hours	1–2 hours	Never	No
3–4 hours	Above 90%	5–6 hours	3–4 hours	Always	Yes

Sample dataset.

4. Quality Issues:

- Imbalance: More “Pass” than “Fail”
- Duplicates: Some students recorded more than once.
- Missing Value: Some students did not answer all questions

5. Learning Type:

- This is **Supervised Learning** problem

Reason: We have a clear label (Pass or Fail). Machine learns from the input features to predict the outcome.

6. Use Case

- ✓ Machine Learning: This dataset can be used for classification.

Predict if a student “Pass” or “Fail”

Schools can identify at-risk students early.

Teachers can help weak students.

- ✓ Data Science Lifecycle:

1. Problem Defining: What factors affects student success.
2. Data Collection: Questionnaire and School records
3. Data Cleaning: Fixing missing values and Errors
4. Model Building: Train a classification model
5. Evaluation: Check accuracy of predictions
6. Deployment: Use model in schools